

Kovvuru Saicharanreddy

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PROFESSIONAL SUMMARY

- **Experienced Data Analyst with 5+ years of experience** in SQL, Python, and BI tools (Power BI, Tableau, Looker). Proven track record in driving business intelligence, **optimizing data workflows**, and developing predictive analytics solutions to enhance decision-making and revenue growth.
- Proven ability to **optimize big data workflows**, drive strategic **risk management**, and **enhance decision-making** through advanced data analytics.
- Adept at leveraging **machine learning models, Databricks, and cloud platforms (AWS, Azure, GCP)** to process **large-scale datasets** for fraud detection, customer segmentation, **revenue forecasting**, and operational **efficiency improvements**.
- Skilled in the complete **Software Development Life Cycle (SDLC)** from requirements analysis to deployment and issue resolution.
- Proficient in building, troubleshooting, and optimizing reporting content, with expertise in **Google Analytics** and Adobe Analytics for digital analytics transformation.
- Specialized in Data Governance, Data Integration, Data Migration, and Metadata Management across diverse domains, including property and casualty insurance.
- **Extensive experience in SQL, PL/SQL, and ETL tools** for **data extraction, transformation, and loading (ETL/DTS)** from diverse databases, including **Oracle, DB2, Sybase, MS SQL Server, and Snowflake**.
- Hands-on experience with **Data Profiling, Data Wrangling, and metadata management**, enhancing **data quality and governance** through the creation of **business glossaries, data dictionaries, lineage documentation, and master data management (MDM) frameworks**.
- Expertise in developing and managing SSIS Packages for efficient data migration, including transforming DTS packages into SSIS for streamlined workflows.
- Proficient in **Big Data technologies** including **Hadoop, Apache Spark, Spark Streaming, and cloud services such as AWS EC2, EMR, Lambda, and RDS**, enabling **real-time data processing** and large-scale analytics.
- Strong analytical skills in developing and executing **SQL-based analytics, data visualization, statistical modeling, and automation**, delivering insights that drive **customer segmentation, churn prediction, and financial risk assessments** for business stakeholders.

TECHNICAL SKILLS

Data Visualization and Analytics Tools	Power BI, Tableau, Looker, Qlik Sense, MS Excel, MS Access, Business Intelligence (BI), KPI Dashboards, Ad Hoc Reporting, Google Analytics
Programming Languages	SQL (Query Optimization, Performance Tuning), Python, PySpark, R, C++, GIT (Version Control), DBT (Data Build Tool for SQL transformations)
Cloud & Big Data Platforms	AWS (EMR, EC2, S3, Athena, Redshift), Azure (Blob Storage, HD Insights, Synapse Analytics, MS SQL Server), Google Cloud (BigQuery, Google Analytics), Snowflake, Databricks, Delta Lake.
Data Analysis Libraries	Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn, ggplot2
ETL & Workflow Automation	Apache Airflow, Prefect, Apache Kafka, UI Path, Selenium with Python, Power Automate, AWS Glue.

Machine Learning & Predictive Analytics	Scikit-Learn, XGBoost, Feature Engineering, Forecasting, Time Series Analysis (ARIMA, Prophet, LSTM), Bayesian Statistics, Hypothesis Testing.
Data Science & Statistical Modeling	Data Exploration, Data Mining, Classification, Clustering, Predictive Analytics, Risk Modeling.
Data Engineering & Warehousing	Apache Spark, Hadoop, Data Warehousing, Data Cleansing, ETL Pipelines, Databricks.
Business & Strategic Analytics	A/B Testing, Customer Segmentation, Revenue Optimization, Churn Prediction, Executive Reporting, Data-Driven Decision Making
Data Governance & Compliance	Data Quality Management (DQM), Metadata Management, Master Data Management (MDM), Data Cataloging, Data Lineage Tracking, GDPR, CCPA, HIPAA Compliance
Communication and Collaboration Skills	Cross-Functional Collaboration, Stakeholder Management, Strategic Decision-Making, Project Management, Agile Methodologies, Data Storytelling

PROFESSIONAL EXPERIENCE

Fifth Third Bank (July 2023 - Present) || Data Analyst | Retail Banking Analytics | United States

Project Description: The Retail Banking Analytics Project enhances fraud detection, credit risk assessment, and customer segmentation through data-driven insights. It focuses on predictive modeling, automated data workflows, and interactive dashboards to support real-time transaction monitoring and risk scoring. By integrating ETL pipelines, CI/CD automation, and KPI tracking, the project ensures efficient, secure, and compliant financial data management.

- Automated **Power BI dashboards** to track key performance indicators (**KPIs**) and monitor customer engagement trends, resulting in an 8% reduction in customer churn by enabling data-driven strategies.
- Built **Python-based machine learning models** to improve the accuracy of credit risk assessments by analyzing borrower profiles, which led to a 5% reduction in loan default rates.
- Designed an **Excel-based pricing** tool by integrating market trends and customer profiles, **leading to a 3% increase** in net interest margin through more informed interest rate recommendations.
- Led **A/B testing** initiatives to assess the effectiveness of marketing campaigns, driving a 7% increase in customer **acquisition through targeted** promotional strategies.
- Optimized **Databricks ETL pipelines** with **Kafka**, reducing query times by **50%** and enhancing **real-time data processing**.
- **Integrated real-time analytics** into banking systems, analyzing lending data across **5+ major product lines**, improving **loan** approval accuracy & risk assessments.
- Directed **data governance** initiatives focused on improving data quality, compliance, and regulatory adherence, ensuring consistent data lineage across retail banking operations.
- Implemented scalable **CI/CD pipelines on Azure using GitLab, Docker, and Kubernetes**, improving deployment frequency by 40%, which enhanced the efficiency of data engineering workflows and reduced manual interventions.

- Managed end-to-end development of **data analytics projects**, collaborating with **engineering, finance, and risk management teams** to enhance business intelligence insights.
- **Enhanced risk modeling strategies** by implementing **Python-driven statistical analyses**, improving **credit risk predictions and reducing loan delinquencies by 6%**.
- Developed **CLV models to assess** customer profitability, optimizing retention strategies and increasing loyalty by 10%.

Environment: SQL, Python, Power BI, Tableau, PySpark, R, AWS (S3, EC2, Redshift), Azure (Synapse, Blob Storage), Snowflake, Apache Spark, Databricks, PostgreSQL, Apache Airflow, AWS Glue, Prefect, CI/CD (GitLab, Docker, Kubernetes), Scikit-learn, Time Series Forecasting (ARIMA, LSTM), A/B Testing, Customer Segmentation, Fraud Detection, KPI Tracking, Risk Modeling.

Infosys Limited (Aug 2021 – Dec 2022) | Data Analyst | Healthcare analytics | India

Project Description: The Healthcare Data Analytics Project streamlined EHR data processing, transformation, and analysis to enhance data quality, compliance, and clinical decision-making. It involved automating ETL pipelines, real-time data processing, and advanced reporting to improve regulatory adherence (HIPAA, SOX, MIPS). By leveraging cloud-based storage, SQL-based data wrangling, and statistical modeling, the project optimized compliance tracking and delivered actionable insights for patient care and healthcare analytics.

- Built **Tableau dashboards** tracking **EHR (Electronic Health Records) data**, improving **operational efficiency by 25%** and reducing **manual** reporting errors.
- Assessed **SQL-based data wrangling & transformation** on healthcare claims, reducing **reporting errors by 30%** through automated **data validation** scripts.
- Normalized and transformed data elements from **Electronic Health Records (EHRs)** and practice management systems to align with a unified data model, supporting **quality improvement initiatives, clinical trials, and real-world evidence studies**.
- Performed **in-depth analysis** on EHR data stored in **AWS S3 buckets** using tools like **AWS Athena, Databricks, and PySpark**, uncovering critical insights and enhancing overall **data quality** for healthcare reporting.
- Developed robust **ETL pipelines using PySpark & Airflow**, enabling seamless data integration workflows across **multi-source healthcare systems**.
- Initiated troubleshooting processes for **ETL failures using AWS Glue & Step Functions**, resolving **data pipeline issues** and minimizing downtime for analytics workflows.
- **Ensured compliance with HIPAA, SOX, and regulatory frameworks**, maintaining **data security and audit readiness** for protected patient records.
- Implemented rigorous **data quality checks** to validate the reliability and effectiveness of data for **Merit-Based Incentive Payment System (MIPS) reporting**, increasing compliance accuracy by **30%**.
- Generated advanced Tableau reports for Year-to-Date (YTD) and Month-End Reporting, integrating **SAS-based statistical models** to **improve patient care decision-making**.
- Wrote and optimized **complex SQL queries** on **PostgreSQL databases** to monitor file pull status, deployment mappings, and maintain **data integrity** across in-house healthcare applications.
- **Collaborated with data engineers, compliance teams, and healthcare analysts** to align **data models** with evolving business requirements, optimizing operational efficiency.
- Orchestrated data pipelines using **Airflow DAGs & custom mapping scripts**, reducing **manual intervention & improving data consistency** across healthcare analytics teams.

Environment: SQL, Python, Tableau, Power BI, AWS (S3, Athena, Redshift), Databricks, Apache Spark, PostgreSQL, PySpark, Apache Airflow, AWS Glue, Prefect, CI/CD (GitLab, Docker, Kubernetes), Scikit-learn,

SAS, Data Wrangling, Risk Modeling, Regulatory Compliance (HIPAA, SOX, MIPS), ETL Pipelines, Data Governance, Data Lineage, and Healthcare Analytics.

ACS Technologies Limited | (Aug 2019 – Jun 2021) | Junior Data Analyst | India

- **Developed interactive dashboards in Power BI & SQL**, monitoring **KPIs for operational performance**, improving data-driven decision-making.
- Optimized **ad hoc business reporting** using **MS Access & SQL**, reducing **manual data retrieval time by 30%**.
- **Engineered OLTP system architectures** to enhance transaction processing speed, improving **data retrieval times by 40%**.
- Designed **SQL-driven ETL pipelines**, improving **data migration accuracy** and reducing **reporting errors** in legacy system integrations.
- **Ensured adherence to internal data security policies**, maintaining **consistent** data integrity across business reporting systems.
- Refined data migration and cleansing processes **during ETL pipeline development**, increasing data accuracy and supporting efficient data-driven decision-making.
- Streamlined **repetitive workflows** using **Unix scripting**, decreasing **manual processing effort by 40%** and increasing **task execution efficiency**.
- **Implemented** data cleansing processes **for** database normalization, improving **data accuracy for operational reports**.

Key Achievements

- **Streamlined ETL workflows** and optimized data pipelines using **Apache Airflow and SQL**, reducing manual reporting time **by 50%** and improving real-time analytics for business intelligence.
- **Engineered predictive analytics models in Python and SQL**, improving **fraud detection accuracy by 25%** and reducing financial **risk exposure in banking and payment systems**.
- Spearheaded and implemented **KPI-driven dashboards using Power BI and Tableau**, reducing manual data processing **efforts by 40%** and enabling real-time decision-making for executives.
- Orchestrated cross-functional **collaboration with data engineers and business teams** to migrate and clean **80TB** of structured and unstructured data **in AWS and Snowflake**, reducing data retrieval **times by 35%**.